TASK ONE:

Using the employees table from the Classic Models database write each employee and the employee they report to. Who is the CEO of the company? How do you know that?

- The CEO must be Diane Murphy (1002), the employee who reports to no one.

```
1 • select
2       concat(e1.firstName, ' ', e1.lastName) as 'Employee', e1.employeeNumber,
3       concat(e2.firstName, ' ', e2.lastName) as 'Manager', e2.employeeNumber, e1.reportsTo
4   from employees as e1
5   left join employees as e2 on e2.employeeNumber = e1.reportsTo
6   order by concat(e1.firstName, ' ', e1.lastName);
```

Employee	employeeNumber	Manager	employeeNumber	reportsTo
Andy Fixter	1611	William Patterson	1088	1088
Anthony Bow	1143	Mary Patterson	1056	1056
Barry Jones	1504	Gerard Bondur	1102	1102
Diane Murphy	1002	NULL	NULL	NULL
Foon Yue Tseng	1286	Anthony Bow	1143	1143
George Vanauf	1323	Anthony Bow	1143	1143
Gerard Bondur	1102	Mary Patterson	1056	1056
Gerard Hernandez	1370	Gerard Bondur	1102	1102
Jeff Firrelli	1076	Diane Murphy	1002	1002
Julie Firrelli	1188	Anthony Bow	1143	1143
Larry Bott	1501	Gerard Bondur	1102	1102
Leslie Jennings	1165	Anthony Bow	1143	1143
Leslie Thompson	1166	Anthony Bow	1143	1143
Loui Bondur	1337	Gerard Bondur	1102	1102
Mami Nishi	1621	Mary Patterson	1056	1056

TASK TWO:

```
1   alter table creature
2   add column worships int,
3   add constraint fk foreign key (worships)
4   references creature (creatureId);
```

```
1 • describe creature;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
creatureId	int	NO	PRI	NULL	
creatureName	varchar(30)	YES		NULL	
creatureTypeID	int	YES	MUL	NULL	
townID	int	YES	MUL	NULL	
worships	int	YES	MUL	NULL	

creatureId	creatureName	creatureTypeID	townID	worships
1	Bannon	1	1	NULL
2	Myers	1	2	NULL
3	Neff	1	3	NULL
4	Neff	1	4	NULL
5	Mieska	1	5	NULL
6	Carlis	1	1	NULL
7	Kermit	2	6	NULL
8	Godzilla	3	7	NULL
9	Thor	4	8	NULL
10	The Tick	4	9	NULL
11	Chairface Ch...	5	9	NULL
12	Arthur	6	9	NULL

Not sure why this is showing a null row at the bottom, it doesn't show up in the CLI.

```
update creature set worships = "The Tick" where creatureName = "Arthur";
update creature set worships = "Kermit" where creatureName = "Godzilla";
update creature set worships = "American Maid" where creatureName = "Bat Manuel";
```

```
update creature set worships = 10 where creatureID = 12;
update creature set worships = 7 where creatureID = 8;
select * from creature;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

creatureId	creatureName	creatureTypeID	townID	worships
1	Bannon	1	1	NULL
2	Myers	1	2	NULL
3	Neff	1	3	NULL
4	Neff	1	4	NULL
5	Mieska	1	5	NULL
6	Carlis	1	1	NULL
7	Kermit	2	6	NULL
8	Godzilla	3	7	7
9	Thor	4	8	NULL
10	The Tick	4	9	NULL
11	Chairface Ch...	5	9	NULL
12	Arthur	6	9	10

1. Write a query that shows creature name and the creature name of their idol. Do not show any creatures who do not have an idol.

```
1 • select c1.creatureName as "Worshipper", c2. creatureName as "Idol"
2   from creature as c1
3   join creature as c2 on c2.creatureID = c1.worships;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Worshipper	Idol
Godzilla	Kermit
Arthur	The Tick

2. Write a query that shows all creatures that do not have an idol.

```
1 • select * from creature where worships is null;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

creatureId	creatureName	creatureTypeID	townID	worships
1	Bannon	1	1	NULL
2	Myers	1	2	NULL
3	Neff	1	3	NULL
4	Neff	1	4	NULL
5	Mieska	1	5	NULL
6	Carlis	1	1	NULL
7	Kermit	2	6	NULL
9	Thor	4	8	NULL
10	The Tick	4	9	NULL
11	Chairface Chippendale	5	9	NULL
NULL	NULL	NULL	NULL	NULL

Aggregation in SQL:

```sql
-- 1. Show each creature and a count of the weapons they possess in decreasing order.
select creatureName as "Name", count(weaponID) as "Weapon #"
from creature as c
join creature_weapon as cw on c.creatureId = cw.creatureId
group by cw.creatureID
order by count(weaponID) desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Weapon #
Kermit	2
Bannon	1
Thor	1

```sql
-- 2. Show the creature that has the most worshipers.
select idol.creatureName, COUNT(worshipper.creatureId) as worshipper_count
from creature as worshipper
join creature as idol on worshipper.worships = idol.creatureId
group by idol.creatureName
having count(worshipper.creatureId) = (
    select max(worshippers)
    from (select count(*) as worshippers from creature
    where worships is not null -- is not null from DeepSeek
    group by worships) counts);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureName	worshipper_count
Godzilla	3

```sql
-- 3. Show the number of creatures that are not worshiped.
select count(*) from (
    select creatureId, creatureName from creature idol
    where not exists (
        select * from creature worshipper
        where worshipper.worships = idol.creatureId)
) as creatures_not_worshipped;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

count(*)
9

```sql
-- 4. Show the most popular weapon.
select cw.weaponID, count(cw.weaponID) weapons from weapon as w
join creature_weapon as cw on w.weaponID = cw.creatureID
group by cw.weaponID
having count(cw.weaponID) = (
select max(weapons) from(
    select count(weaponID) as weapons from creature_weapon
    group by weaponID) counts);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

weaponID	weapons
1	4

```sql
-- 5. Show the town that has the most villains.
select town.townName, count(creature.townID) from creaturetype
join creature on creature.creatureTypeID = creaturetype.creatureTypeID
join town on creature.townID = town.townID
where creature.creatureTypeID = 5
group by town.townName
order by count(creature.townID) desc -- DeepSeek said this was the most efficient
limit 1;                             -- way to return the highest number
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

townName	count(creature.townID)
The City	2